AE 481 Team 01 Assignment 8

November 10, 2025

1 Drag Polars

The estimation of drag polars for the refinement of sizing and design trades is discussed in this section.

1.1 Subsonic Parasite Drag

The component buildup method from Raymer [1] is used to estimate the subsonic parasite drag. Use of AVL [2] was deferred until after the sizing refinement, as it would yield inaccurate results due to differences in wing area and airfoil. An updated model using AVL will be implemented in future assignments.

The subsonic parasite drag is estimated using equation 1. The components considered in the parasite drag buildup are the wing, fuselage, canopy, horizontal and vertical tails, drop tanks, and pylons.

$$(C_{D_0})_{\text{subsonic}} = \frac{\sum(C_{f_c}\, FF_c\, Q_c\, S_{\text{wet}_c})}{S_{\text{ref}}} + C_{D_{\text{misc}}} + C_{D_{\text{L\&P}}} \tag{1}$$

1.1.1 Skin Friction

The equivalent flat-plate skin friction coefficient C_f of each component is estimated using equation 2. It is assumed that the entirety of the skin has turbulent flow, as the Reynolds number is high. This assumption is also more conservative than assuming any percentage of laminar flow.

$$C_f = \frac{0.455}{log_{10}(R)^{2.58}(1 + 0.144M^2)^{0.65}} \tag{2}$$

The form factor of each of the components is estimated using equations 3, 4, and 5, which come from Raymer. f is the fineness ratio of the body, expressed in equation 6. For the vertical tail, the FF is multiplied by 1.1 to account for the hinge line.

Wing, tail, strut, and pylon:

$$FF = \left[1 + \frac{0.6}{(x/c)_m}\left(\frac{t}{c}\right) + 100\left(\frac{t}{c}\right)^4\right][1.34M^{0.18}(\cos\Lambda_m)^{0.28}] \tag{3}$$

Fuselage and canopy:

$$FF = \left(0.9 + \frac{5}{f^{1.5}} + \frac{f}{400}\right) \tag{4}$$

Nacelle and external store:

$$FF = 1 + (0.35/f) \tag{5}$$

Fineness Ratio:

$$f = \frac{l}{d} = \frac{l}{\sqrt{(4/\pi)A_{max}}} \tag{6}$$

The interference factor Q for each component is listed in Raymer, and summarized in Table 1. Favorable interferences are ignored at this stage.

Table 1: Interference Factors

Component	Q
Fuselage	1
Wing	1
Tail	1.05
Drop Tanks	1.4

1.1.2 Miscellaneous, Leaks and Protuberances

The components included in the $C_{D,misc}$ term are the wheels, landing gear struts, and arresting hook. For the wheels and struts Raymer lists values of C_{D_π}, which is defined as the D/q divided by the frontal area.

For a combined wheel and tire, the C_{D_π} is 0.25. For a strut with Re $> 3 \times 10^5$, the C_{D_π} is 0.3. Interference is accounted for by multiplying by 1.2, and an additional 7% is added to account for landing gear doors.

For a USN arresting hook, Raymer lists D/q as 0.15 ft^2.

To account for leakage and protuberances such as antennas and panel edges, the $C_{D_{L\&P}}$ is approximated as 5% of the total drag, which is the upper bound of the stealth fighter estimate listed in Raymer.

1.2 Supersonic Parasite Drag

The supersonic parasite drag is estimated using equation 7, which is similar to the subsonic case discussed in §1.1.

$$(C_{D_0})_{\text{supersonic}} = \frac{\sum(C_{f_c}\, S_{\text{wet}_c})}{S_{\text{ref}}} + C_{D_{\text{misc}}} + C_{D_{\text{L\&P}}} + C_{D_{\text{wave}}} \tag{7}$$

In the skin friction calculation, the C_f is the same as in §1.1.1. The interference factor is not included, as it is accounted for in the $C_{D_{wave}}$ term. Thus, FF and Q are both unity. The $C_{D_{\text{misc}}}$ and $C_{D_{\text{L\&P}}}$ are calculated as in §1.1.2.

1.2.1 Wave Drag, $M > 1.2$

For Mach numbers above 1.2, the wave drag can be calculated using equation 8. The original value for the middle 0.2 term was 0.386, but it was changed to 0.2 based on advice from Raymer.

$$(D/q)_{\text{wave}} = E_{\text{WD}} \left[1 - 0.2\,(M - 1.2)^{0.57} \left(1 - \frac{\pi\,\Lambda_{LE\text{-deg}}^{0.77}}{100} \right) \right] (D/q)_{\text{Sears–Haack}} \tag{8}$$

The E_{WD} is the wave drag efficiency factor, and is estimated as 2.2. This is the upper bound of the historical values listed in Raymer for typical supersonic fighters.

The D/q of an equivalent Sears–Haack body, is calculated using equation 9. The A_{max} is the maximum cross-sectional area of the aircraft. If the location of maximum cross-sectional area $\ell_{A_{\text{max}}}$ is well behind the halfway point of the fuselage, the ℓ is defined as double the $\ell_{A_{\text{max}}}$.

$$(D/q)_{\text{Sears–Haack}} = \frac{9\pi}{2} \left(\frac{A_{\text{max}}}{\ell} \right)^2 \tag{9}$$

1.2.2 Wave Drag, $M_{crit} < M < 1.2$

The wave drag in the transonic regime is calculated using the graphical method described in Raymer. The M_{crit}, M_{DD}, and associated drag rise $C_{D_{\text{wave},DD}}$ are calculated using the Korn equation described in the Metabook [3]. The $C_{D_{\text{wave}}}$ at $M = 1.2$ is also found from equation 8. The $C_{D_{\text{wave}}}$ at $M = 1.05$ is assumed to be the same as at $M = 1.2$. The $C_{D_{\text{wave}}}$ at $M = 1$ is assumed to be half of that at $M = 1.05$ and $M = 1.2$, and the drag is assumed to vary linearly between $M = 1$ and $M = 1.05$. This gives sufficient fidelity to fit a spline to these points, which smoothly blends between transonic and supersonic drag.

1.3 Drag Due to Lift

The drag due to lift is calculated using historical estimations from Raymer. In future assignments, the drag due to lift will come from AVL.

The subsonic value of K in the drag polar equation is calculated using equation 10, where AR is the wing aspect ratio and e is the Oswald efficiency factor. Supersonic K is calculated using equation 12. This factor accounts for both induced drag due to inviscid effects, and drag caused by flow separation due to lift. e can be estimated using equation 11 from Raymer, which is based on historical aircraft.

$$K = \frac{1}{\pi ARe} \tag{10}$$

$$e = 4.61(1 - 0.045AR^{0.68})(\cos\Lambda_{\text{LE}})^{0.15} - 3.1 \tag{11}$$

$$K = \frac{AR(M^2 - 1)cos\Lambda_{\text{LE}}}{(4AR\sqrt{M^2 - 1}) - 2} \tag{12}$$

1.4 Flap Drag

Raymer provides two equations for calculating flap drag, one for the change in parasitic drag and one for the change in induced drag. The change in parasitic drag due to flaps is calculated using equation 13.

F_{flap} is a constant for plain flaps and slotted flaps equal to 0.0144 and 0.0074, respectively. C_f is the chord length of the flap, with C as the local chord length of the wing. S_{flapped} is the reference of the wing where the wing has a flap. The flap angle, δ_{flap}, is given in degrees.

$$\Delta C_{D_0} = F_{\text{flap}}(\frac{C_f}{C})(\frac{S_{\text{flapped}}}{S_{\text{ref}}})(\delta_{\text{flap}} - 10) \tag{13}$$

Additionally, Raymer discusses a change in induced drag due to flaps. Due to difficulty in estimating the change in C_L from flap deflections, this is accounted for by estimating a Δe, a change in efficiency due to flap deflection. This is estimated using AVL to be -0.046 for full flaps, and -0.032 for half flaps. This was collected at a single C_L, so this will be refined for future assignments. The initial estimates for deflection angles used in the calculation are given in Table 2. NATOPS [4] gives flap deflections for the F/A-18, which gives an estimated for full flaps configuration. However, further work will need to be done to refine the half-flaps configuration estimates.

Table 2: Flap deflection angles used for flap drag calculations

Configuration	Leading Edge Flaps	Trailing Edge Flaps
Full Flaps	15°	35°
Half Flaps	7.5°	17.5°

1.5 Trim Drag

The calculation of the trim drag is difficult as the downwash on the tail must be estimated. For now, the trim drag is estimated as a 5% increase in the drag due to lift. This can be found from a paper by Roskam [5], where a range of the drag due to lift increase from 0.5% to 5% is given. The upper bound of this range is selected due to the large tail area compared to the wing.

1.6 Results

Parasitic drag is estimated for three flight conditions: cruise, takeoff, and approach. During cruise, flaps, landing gear, and the arresting hook are not deployed. For takeoff, flaps are full and landing gear are deployed. For approach, flaps are half and landing gear and the arresting hook are deployed. All flight conditions include 2 external drop tanks which is calculated as skin friction drag. Landing gear and the arresting hook are considered as miscellaneous drag. Flap drag is a fixed increase in C_{D0}. Leaks and protuberances is not considered to be influenced by flap drag. Wave drag is only modeled for the clean aircraft state (no flaps, landing gear, external tanks), and should only be considered for the cruise condition.

Figure 1 displays the breakdown of skin friction, miscellaneous, flap, leakages and protuberance, and wave drag. Half and full flaps cause a large increase in C_{D0}, especially for full flaps. The parasitic drag coefficient nearly doubles compared to the drag at cruise. For the cruise condition, wave drag is the largest contribution to parasitic drag at supersonic speeds. When compared to parasitic drag coefficients listed in Raymer for fighter aircraft, C_{D0} for the F/A-67 is greater by at least 200 counts. This highlights the need for refinement of the aircraft wing and fuselage, to reduce wave drag.

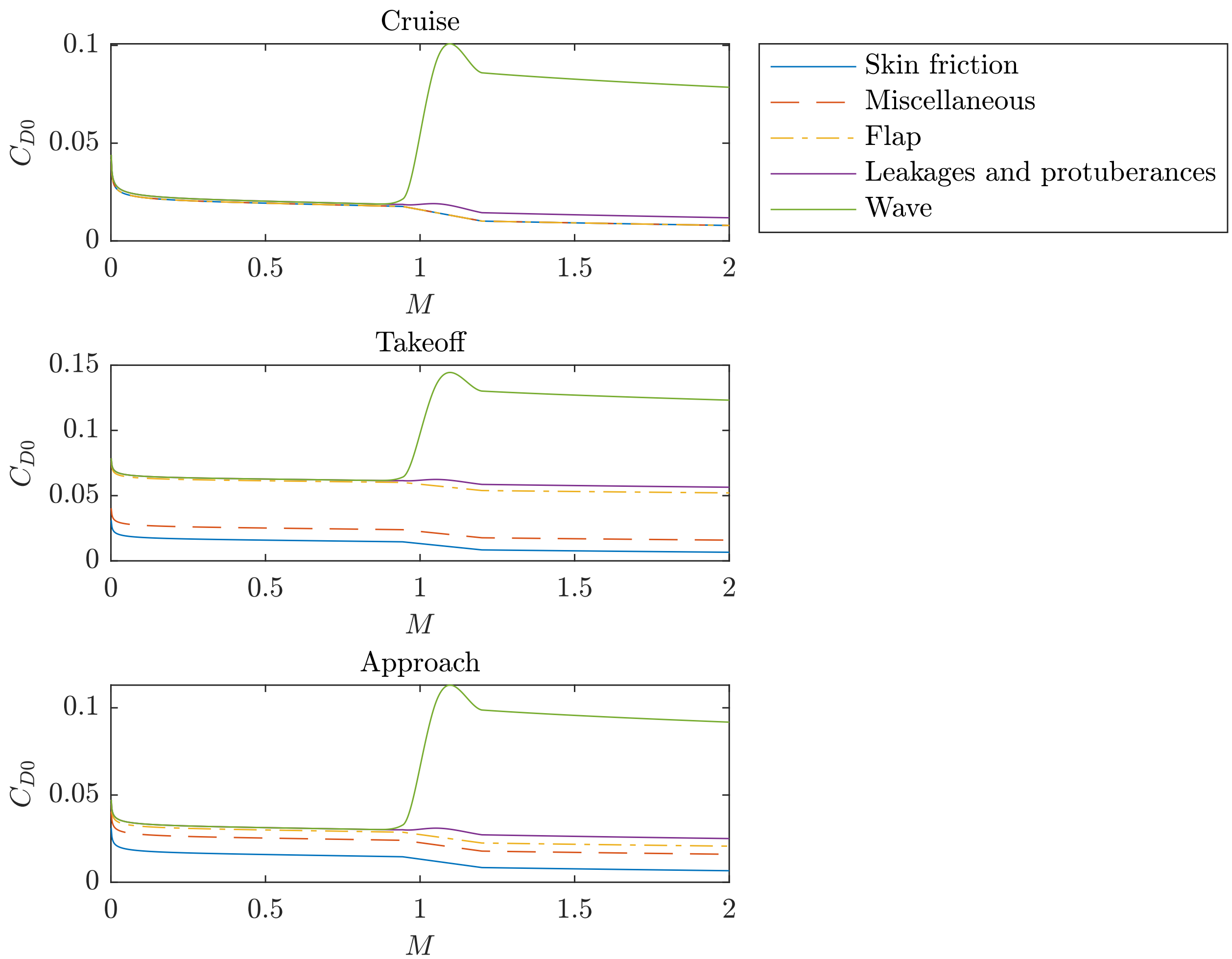

Figure 1: Parasitic drag breakdown for cruise, takeoff, and approach flight conditions.

Drag polars were also generated for the same three flight conditions at their nominal Mach numbers (Fig. 2). This is Mach 0.84 for cruise, and the respective speeds for takeoff and approach. The drag polars show the clear increase in C_{D0} due to flaps and landing gear. The increase in $C_{L,\max}$ due to flaps is also shown and is discussed later. Trim drag is modeled for all conditions as a 5% increase to K.

2 Lift Estimation

2.1 2D $C_{\ell_{\max}}$

Because the F/A-67 uses slotted flaps, it is not possible to model the $C_{\ell_{\max}}$ using XFOIL. A multi-element code such as MSES is required, which is unavailable to the team. Therefore, lower fidelity estimation methods are used.

First, the clean $C_{\ell_{\max}}$ must be found. Because of the wing airfoil's sharp leading edge and high Reynolds number, it cannot be analyzed by XFOIL or ADflow at high angles of attack. Instead, experimental data can be used to approximate the $C_{\ell_{\max}}$. In NACA TN2502 [6], the stall behavior of the NACA 64A006 airfoil

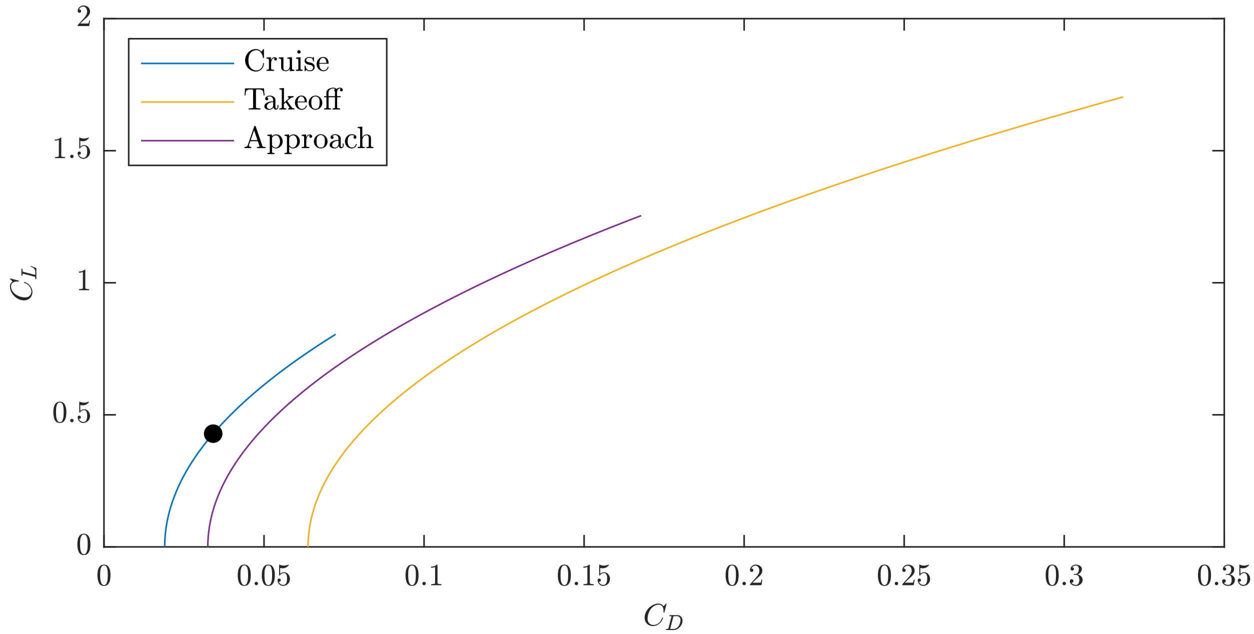

Figure 2: Drag polars for cruise, takeoff, and approach flight conditions. The cruise point at mid-mission weight is marked. Curves terminate at $C_{L,\max}$.

is studied at a Reynolds number of 5.8×10^6. This is assumed to behave similarly to the wing airfoil, as it exhibits leading edge stall. The Reynolds number is also comparable to the takeoff Reynolds number of 17×10^6. It can be seen from Table 3 that the $C_{\ell_{\max}}$ of the NACA 64A006 is 0.8865.

Table 3: Lift Data for NACA 64A006

α	C_ℓ
-0.0459	-0.0011
0.9756	0.1142
1.9542	0.2316
2.9752	0.3556
3.9699	0.4587
4.5014	0.5296
5.0049	0.5629
5.5080	0.6052
5.9810	0.6551
6.9866	0.7533
8.0222	0.8440
9.0599	0.8865
10.0112	0.8642
11.0241	0.7937

To find the flapped $C_{\ell_{\max}}$, empirical adjustments from Raymer are used. These are in the form of $\Delta C_{\ell_{\max}}$ values, which is 0.9 for a plain flap, 1.3 for a slotted flap, and 0.3 for a leading edge flap. There are three different airfoil sections along the wing, listed in Table 4. The associated total $\Delta C_{\ell_{\max}}$ and $C_{\ell_{\max}}$ values are also listed.

Table 4: Airfoil Sections

Section	LE Device	TE Device	$\Delta C_{\ell_{\max}}$	$C_{\ell_{\max}}$
1	None	None	0	0.8865
2	Plain	Slotted	1.6	2.4865
4	Plain	None	0.3	1.1865

2.2 3D $C_{L_{\max}}$

The clean $C_{L_{\max}}$ of the aircraft is estimated using AVL using the critical section method. The C_L is increased until the wing C_ℓ exeeds the clean $C_{\ell_{\max}}$ of 0.8865. This also included tail trim. The clean $C_{L_{\max}}$ is found to be approximately 0.805.

The $C_{L_{\max}}$ can be estimated using equation 14 from Raymer.

$$\Delta C_{L_{\max}} = 0.9 \Delta C_{\ell_{\max}} \left(\frac{S_{\text{flapped}}}{S_{\text{ref}}} \right) \cos \Lambda_{\text{H.L.}} \tag{14}$$

The $\Delta C_{\ell_{\max}}$ for each high-lift device is calculated as 0.1625 for the LE flap, 0.3678 for the TE inboard flap, and 0.3078 for the TE outboard flap. Using these values and the clean $C_{L_{\max}}$ found above, the flapped $C_{L_{\max}}$ is found to be 1.7034.

The predicted $C_{L_{\max}}$ is lower than what was previously used during initial sizing, but they do not make the takeoff, landing, catapult launch, and recovery constraints constraining.

3 Stability

The F/A-67 static margins are within acceptable bounds. As seen in Table 5, the aircraft is positively statically stable in empty and loaded conditions for both missions. According to Raymer, older fighter jets are typically designed to be approximately 5% statically stable, with newer fighter jets designed up to -15% statically stable. Thus, the F/A-67 is within an acceptable range, with room to move the static margin slightly back if needed. If a design change is made such that the static margin becomes negative, also known as relaxed stability, the negative static margin would be compensated for with advanced flight control systems.

The subsonic neutral point prediction comes from AVL, which models the wing, tails, and a cruciform approximation of the fueslage. At supersonic speeds, the neutral point shifts backward. This backwards shift is calculated using AeroSandbox AeroBuildup [7][8], a modified Python implementation of USAF Digital DATCOM [9]. The subsonic neutral point is located 29.4 ft from the nose, and the supersonic neutral point is located 32.5 ft from the nose.

4 Externals Drag Trade Study

External ordnance and fuel tank drag are estimated using the methods discussed in §1.1. The base configuration is the F/A-67 with the fuselage as previously designed with space for internal weapons and fuel carriage. For external store analysis, the aircraft is assumed to have two pylons on each wing with pylons on the wingtip to carry AIM-9X's. Each wing pylon would carry either 2 MK-83 JDAM's or 2 AIM-120C's, with the last 2 AIM-120C's on the fuselage. Raymer gives an interference factor of 1.5 for external stores

Table 5: Static margins for various configurations

Configuration	Subsonic SM	Supersonic SM
Empty	4.5%	29.7%
Air-to-air (empty + payload)	6.5%	31.7%
Strike (empty + payload)	8.1%	33.3%
Air-to-air (no external tanks)	4.6%	29.8%
Strike (no external tanks)	5.7%	31.0%
Air-to-air (fully loaded)	7.9%	33.1%
Strike (fully loaded)	8.9%	34.1%

close to the fuselage, 1.3 for external stores placed more than a diameter away from the fuselage and 1.25 for wingtip-mounted missiles. This analysis includes the pylon drag for each configuration. The results are given in Table 6 for a subsonic cruise condition.

Table 6: Change in drag for external ordnance and fuel tanks, accounting for necessary pylons.

Configuration	ΔC_{D_0}	% increase in Drag
Air-to-Air Ordnance	0.00232	13.7%
Strike Ordnance	0.00242	14.2%
2 External Fuel Tanks	0.00174	10.1%

Adding 2 480-gal external fuel tanks would contribute the least added drag, with the strike ordnance contributing the highest drag. It is clear that internal ordnance and fuel storage will help reduce drag. Currently, the aircraft is capable of holding all weapons and fuel internally. Compared to the potential external drag, the fuselage drag would have to be decreased up to 33% to match the drag of the current aircraft. This is not feasible due to engines taking up a fixed volume and the skin friction remaining relatively the same due to the overall length of the aircraft. The size and form factor of the fuselage would have to be drastically reduced to decrease drag to counteract the additional drag from externals, which is not efficient.

Furthermore, internal carriage is highly desired for supersonic flight due to large increases in drag at higher speeds. It is important to note that, from Raymer graphs, the drag area begins to increase significantly in the transonic region and is not plotted for the supersonic region. This indicates very large values of drag for any external stores, particularly large bombs like the MK-83 JDAM, in the supersonic regime. Therefore, for supersonic flight, it can be assumed that internal storage is vastly preferred due to large increases in drag.

5 Future Refinement

The results of the drag buildup will be used for mission weight estimation and refining sizing plots. The important factor is the modeling of drag divergence, which was not considered in preliminary sizing analyses. This should result in climb and sustained turn speeds that are not supersonic or high transonic, which is unrealistic due to drag divergence. The initial results of the external stores trade will be used to determine the usage of external drop tanks. Additionally, the fuselage shape will be refined to reduce supersonic wave drag.

Although the currently predicted values of $C_{L,\max}$ maintain a design point in the feasible sizing region, further analysis will be considered for resizing the flaps. This may allow for a designs with improved mission performance.

6 Group Work Split

Table 7: Work Split

Person	Work Percentage	Brief Description of Work Done
Michael Chen	18%	Implemented drag buildup code and drag polar, and did report writing.
Charles Choi	10%	Refinement of robust landing gear sizing, static loads, angles, and report writing.
Cristina Erskine	18%	Assisted with drag buildup (specifically flaps and external carriage of ordnance and weapons), stability section and report writing.
James Gold	18%	Assisted with getting areas and other critical dimensions for detailed drag buildup. Assisted with externals drag trade studies.
Santiago Ramos-Assam	18%	Obtained important parameters and cross sectional areas along the length of the aircraft from the CAD for the drag buildup.
Thomas Sheridan	18%	Worked on drag buildup, $C_{L_{\max}}$ estimation, stability, and report writing.

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